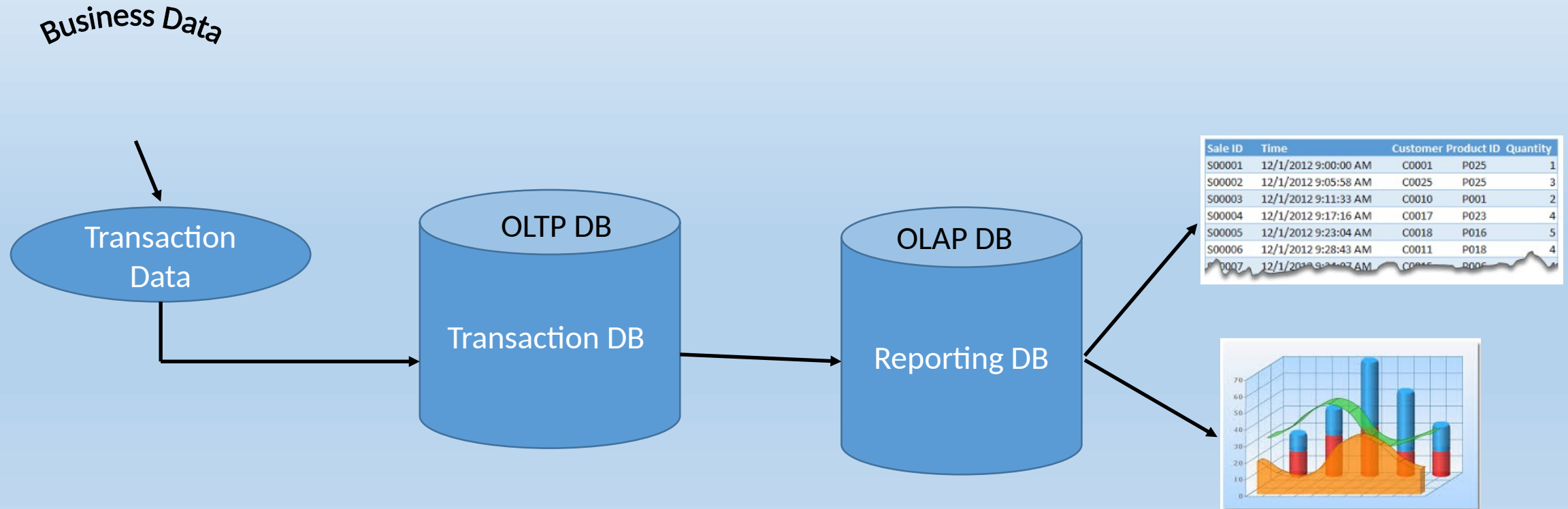


Data Warehousing

Topics :

- OLAP Database.
 - OLTP Database.
 - Evolution of DWH.
 - DWH Architecture.
 - Staging Area.
 - DWH Data base.
 - DWH DB Characters.
 - Data Marts
 - Types of Data Marts.
 - * Dependent Data Mart.
 - * Independent Data Mart.
- 
- ODS
 - Types Approaches
 - * Top –Down Approach.
 - * Bottom – up Approach .
 - ETL Process .
 - ETL Tools
 - Reporting Tools.
 - Granularity.

OLTP & OLAP Database:



OLTP Database:

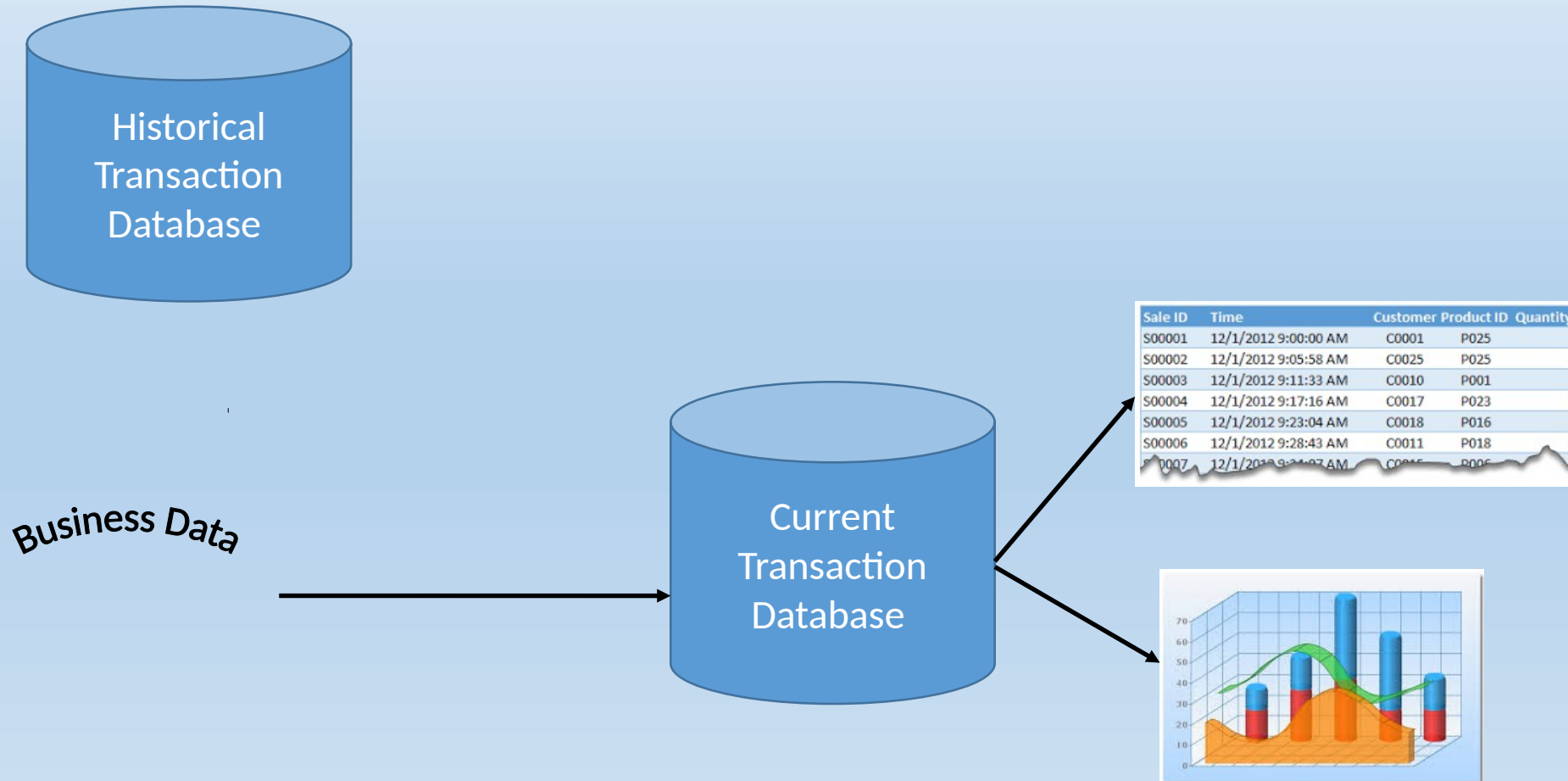
- Online Transaction Processing.
- The databases which are used to store transaction data will be referred as OLTP Database
- OLTP databases contains Transaction Data only.
- OLTP contains Detail Data.
- Generally OLTP databases are designed by using ER modeling
- OLTP databases are highly Normalized.
- Query processing is slow.
- Memory management is Good.

OLAP Database:

- Online Analytical Processing .
- The databases which are used for reporting and analysis is called OLAP databases.
- OLAP databases contains entire organization data.
- OLAP database data is summarized.
- OLAP databases are designed by using Multi Dimensional Modeling.
- OLAP databases are highly De-normalized .
- Query performance is good in OLAP databases.

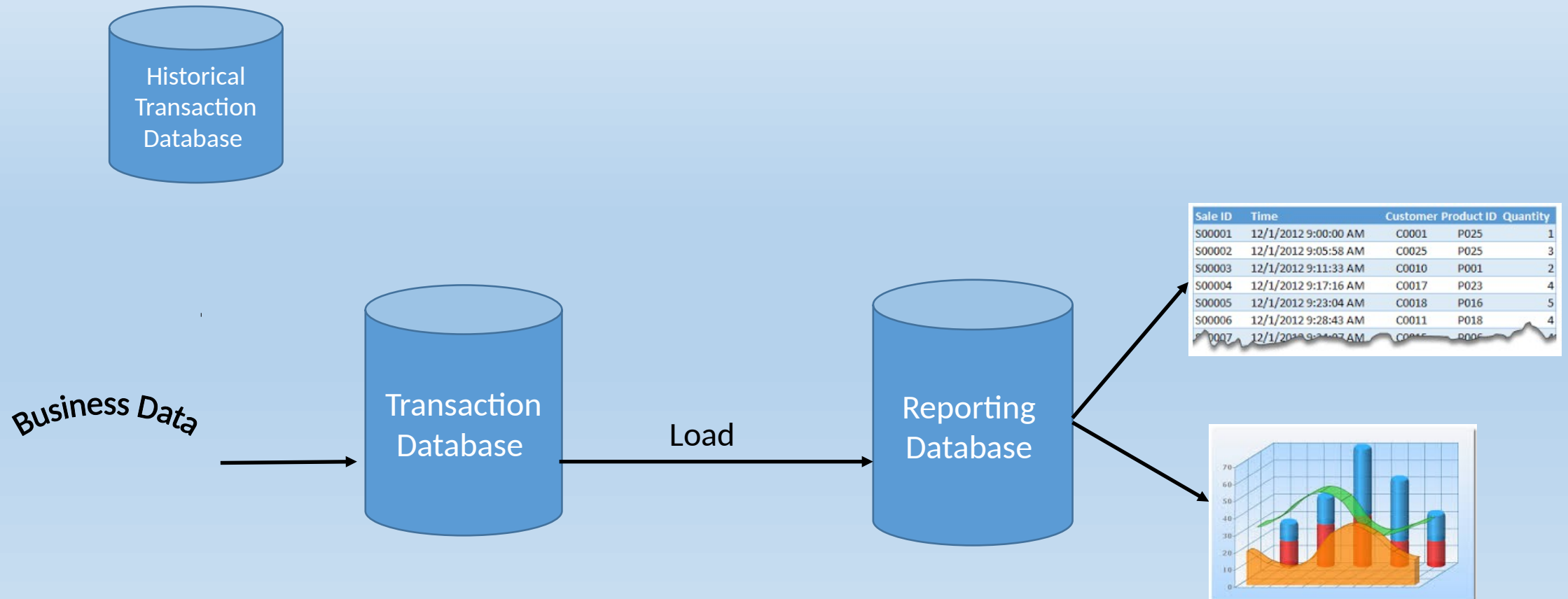
Evolution of DWH

Stage :1



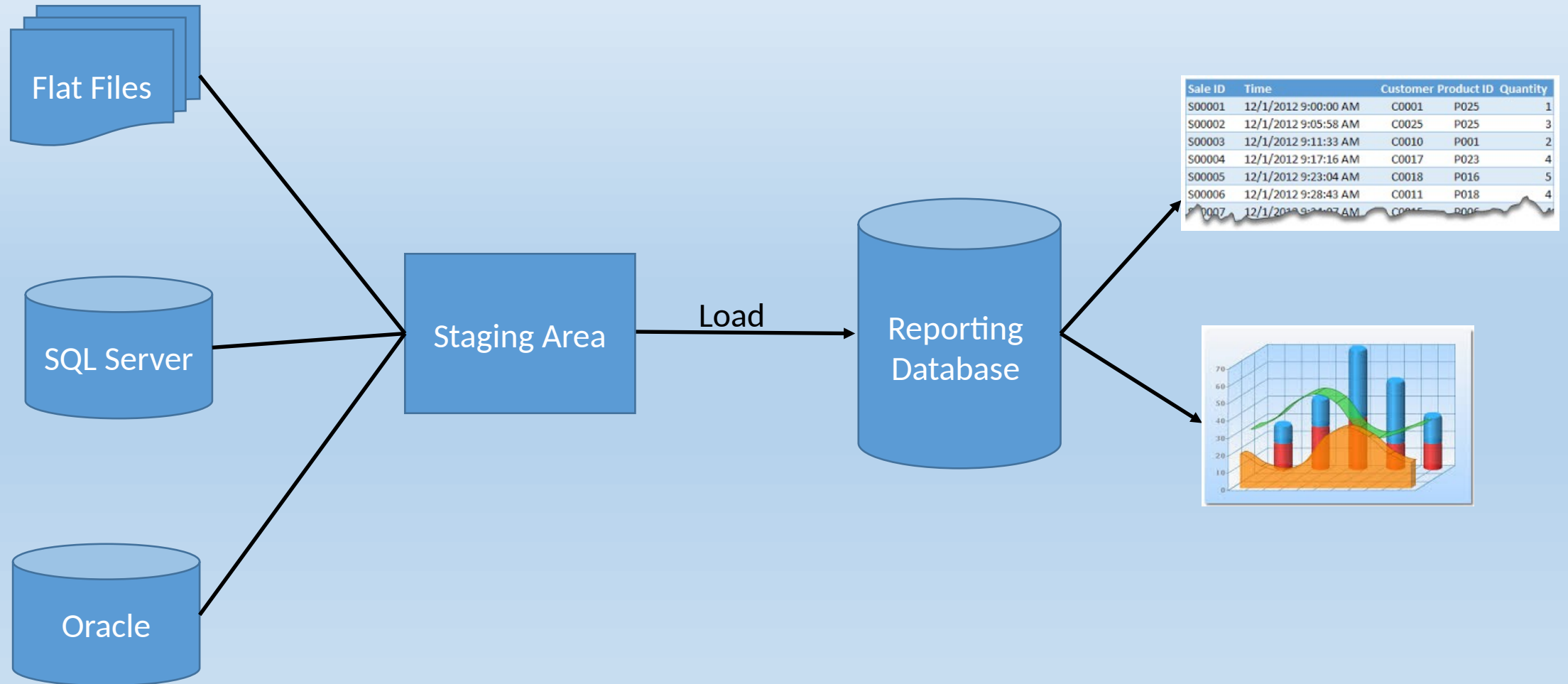
Evolution of DWH

Stage :2



Evolution of DWH

Stage : 3



Evolution of DWH

- Staging Area:

Staging area is a temporary store where we load raw data from different source systems and then we will apply the transformations on the staging data, once data got transformed then we load data in to Reporting Database / OLAP Database / DWH Database.

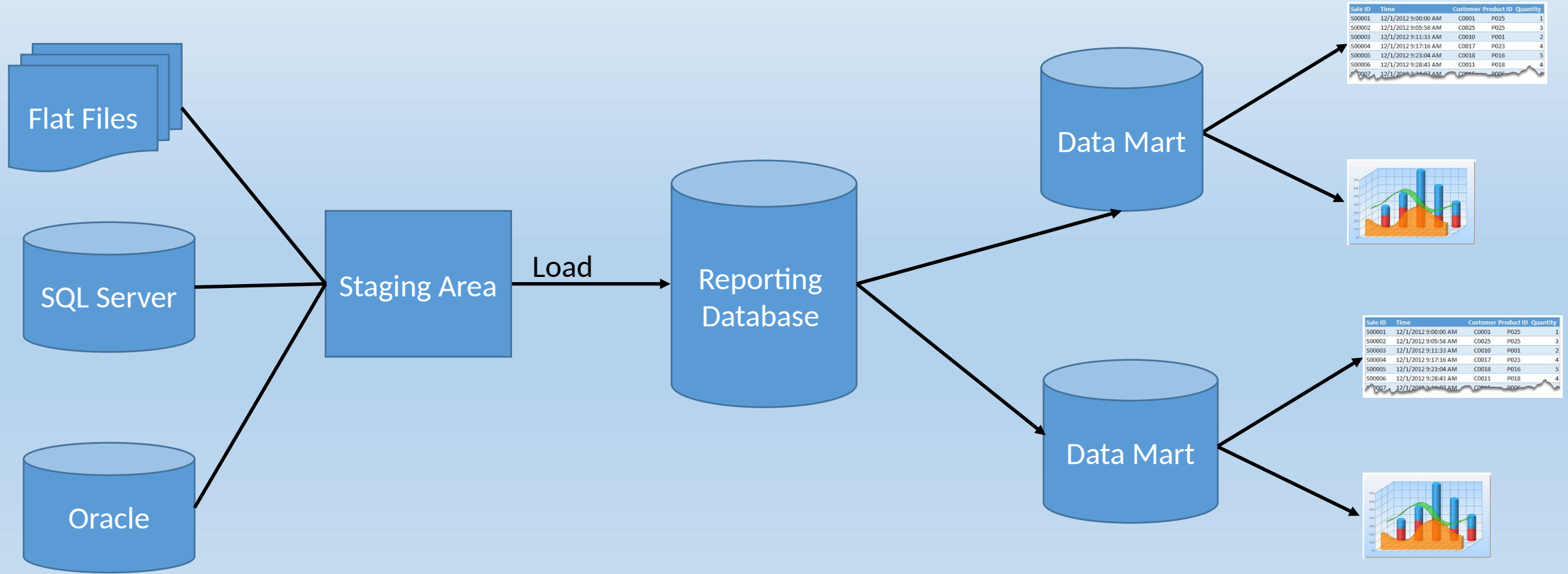
We basically need staging area to hold the data, and perform data cleansing and merging, before loading the data into warehouse.

Data Warehousing

- Data warehousing is a separate database which is specially structured for reporting and analysis .
- Data warehousing contains both current and historical data .
- It is also called as OLAP database or reporting database.
- Generally we will define DWH database by using Multi Dimensional Modeling .

Evolution of DWH

Stage :4



•Data Mart :

- * Data marts are subset of DWH database.
- * Data Marts contains one subject area of information .
- * Data marts are used to improve performance.
- * Creation of data marts are optional in DWH projects.

Types of Data Marts :

1. Dependent Data Mart.
2. Independent Data Mart.

- Types of Data Marts :

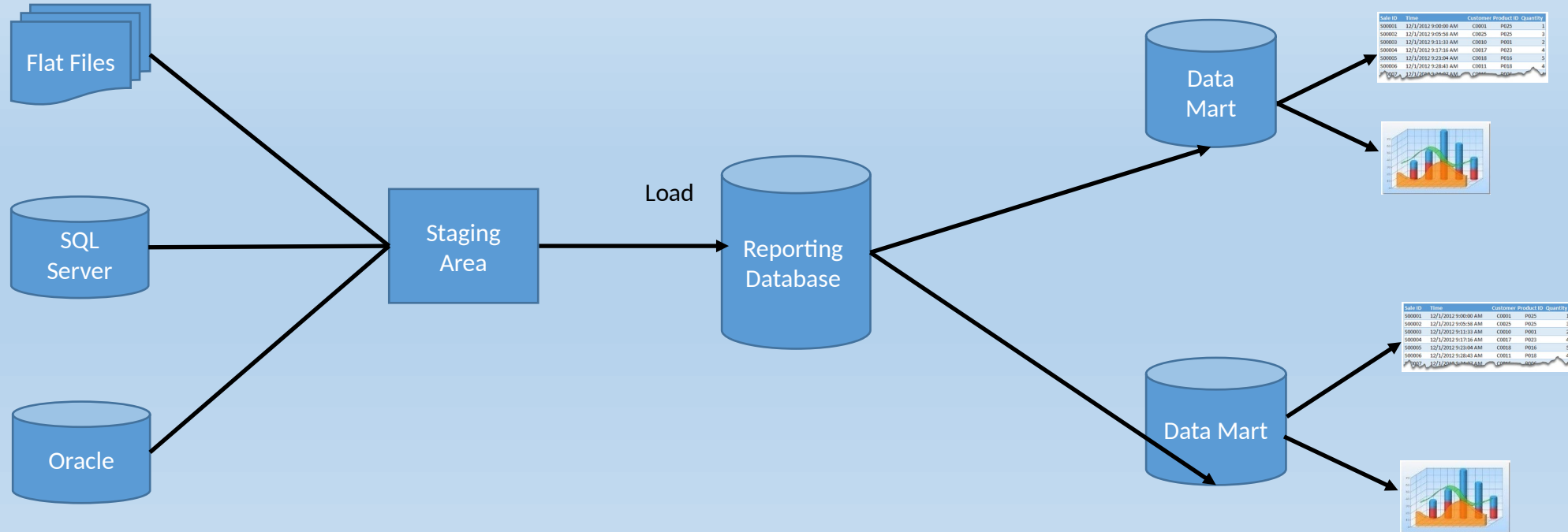
- There are two types of data marts :

- 1. Dependent Data Mart.

- 2. Independent Data Mart.

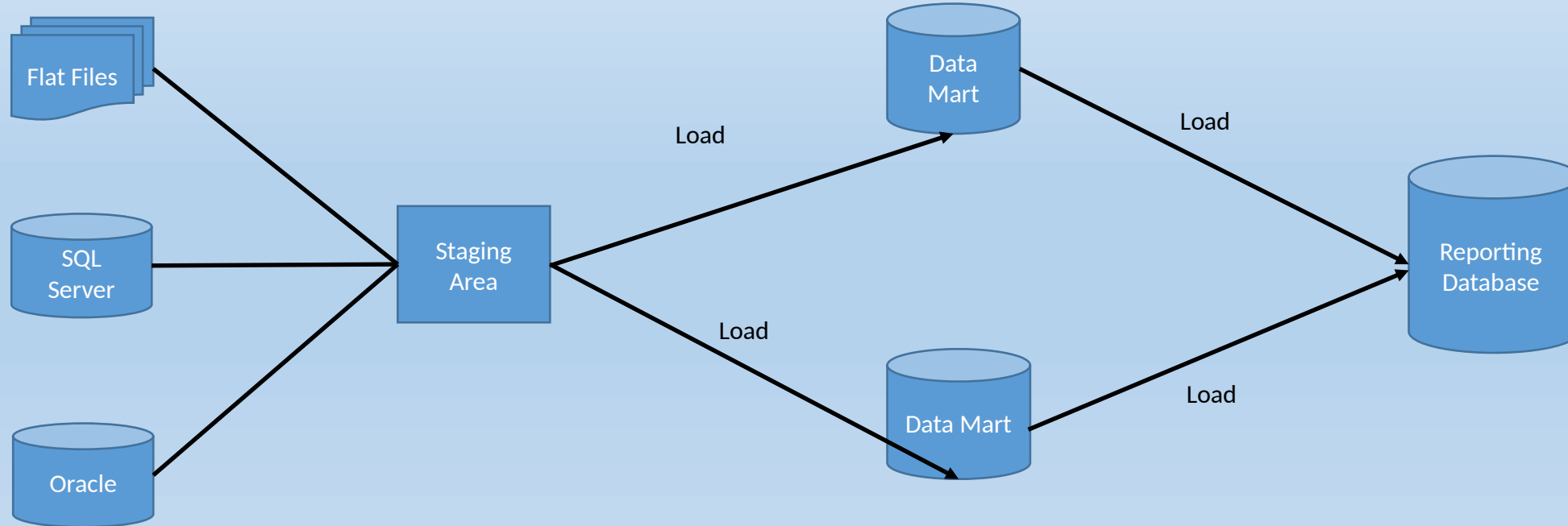
1. Dependent Data Mart.

- * The creation of data mart which depends on DWH is called dependent data mart.
- * you can find this data marts in Top-Down approach.



2. Independent Data Mart.

- * The creation of data mart which does not depends on DWH is called independent data mart .
- * you can find this data marts in Top-Down approach.

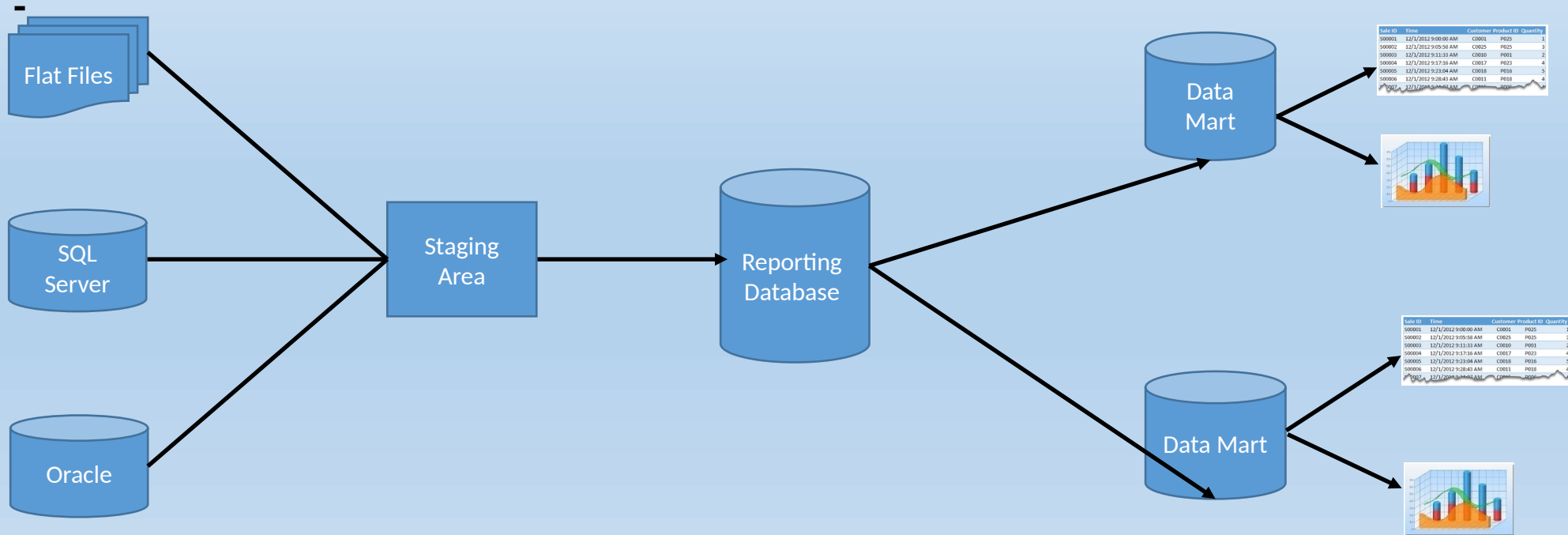


Approach :

1. Top – Down Approach .
2. Bottom – Up Approach.

Top - Down Approach .

- * In the top-down approach the DWH is built first
The data marts are then created from the data warehouse.
- * In this approach we call Data Marts as Dependent Data Mart.



Top – Down Approach .

Advantages :

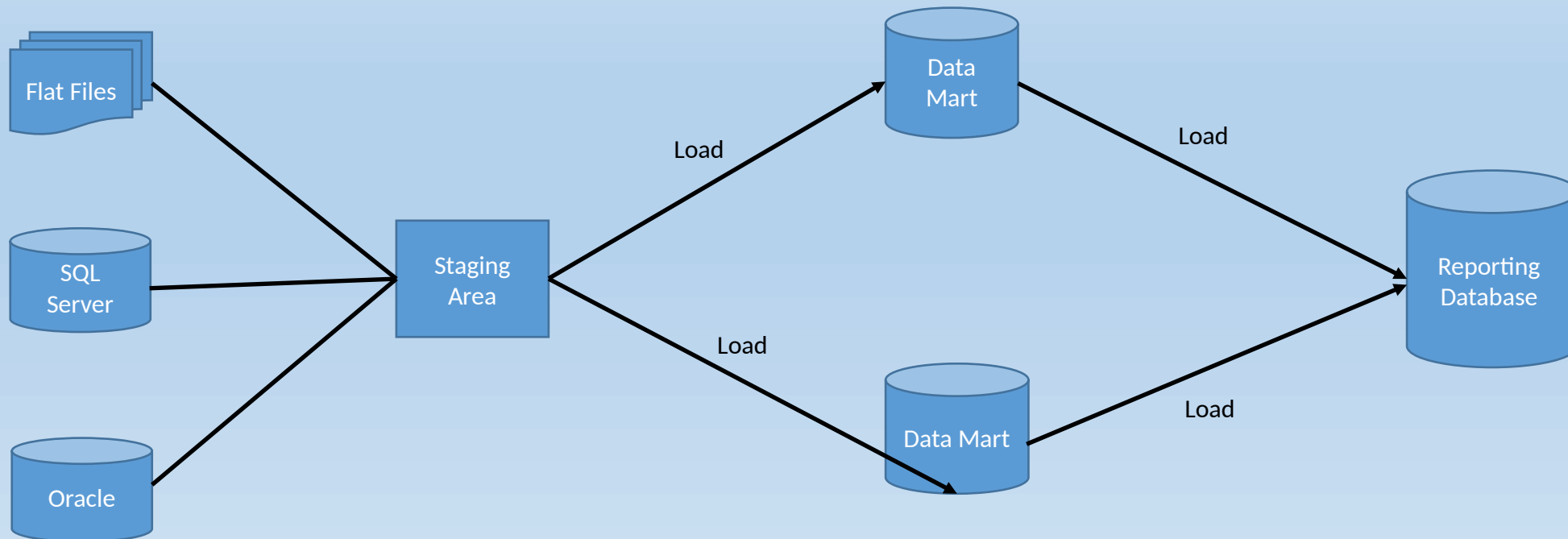
- * Provides consistent dimensional views of data across data marts, as all data marts are loaded from the data warehouse.
- * This approach is robust against business changes.
Creating a new data mart from the data warehouse is very easy.

Disadvantages :

- * This methodology is inflexible to changing departmental needs during implementation phase.
- * It represents a very large project and the cost of implementing the project is significant.

Bottom-Up Approach.

- * In the top-down approach the data marts are created first to provide reporting capability.
- * These data marts are then integrated to build a complete data warehouse.
- * In this approach we call Data Marts as Independent Data Mart.



Bottom-Up Approach.

Advantages :

- * As the data marts are created first, reports can be generated quickly.
- * We can skip in creating the DWH , if there is no integrated reports.

Disadvantages :

- * Integrated reports need to develop after DWH built.

DWH Database Characteristics :

1. Complete Data :

- It contains both current and Historical data.

2. Integrated Data :

- We extract data from different source systems and maintain in the data in a single format in DWH



3. Static Data:

- Once you loaded the data in to the DWH you should not change the data .
- if there is any updated happened on the base database then load that data with flgs

Example : Insert -is- insert / Update -is- Insert

1 Rajesh 5000 -→ 1 Rajesh 5000

1 Rajesh ~~5000~~ -→ 1 Rajesh (-5000)

DWH Database Characteristics :

4. Summarized Data :

Cust_ID	Name	Qty	Amt	Date
C1	Rajesh	2	200	D1
C1	Rajesh	3	300	D4
C1	Rajesh	2	200	D9
C1	Rajesh	1	100	D9
C1	Rajesh	3	300	D20
C1	Rajesh	2	200	D21
C1	Rajesh	1	100	D32
C1	Rajesh	2	200	D32
C1	Rajesh	1	100	D36
C1	Rajesh	1	100	D37



Cust_ID	Name	Qty	Amt	Date
C1	Rajesh	5	500	W1
C1	Rajesh	3	300	W2
C1	Rajesh	5	500	W3
C1	Rajesh	2	200	W5
C1	Rajesh	2	200	W6

D1 -> 1-Jan-2016 (Monday)

D1 - D7 → Week1

D8 - D14 → Week2

D15 - D21 → Week3

D22 - D28 → Week4

D29 - D35 → Week5

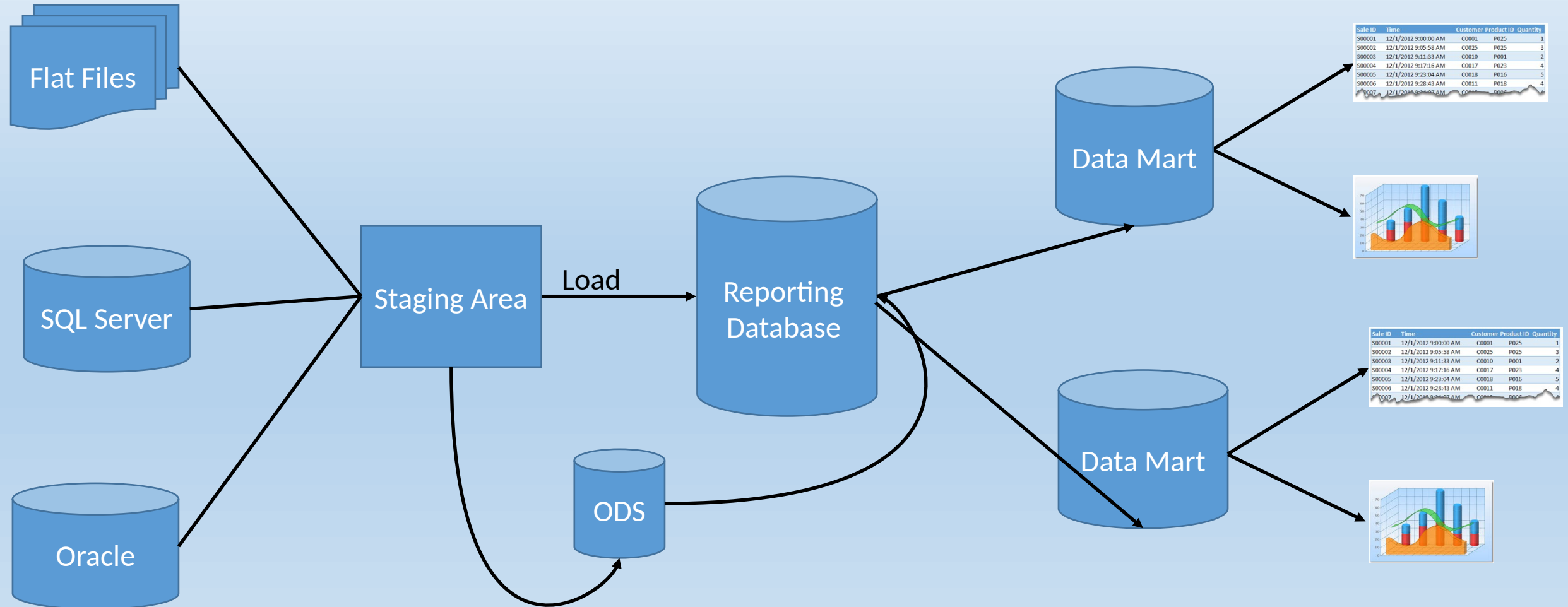
D35 - D42 → Week6

Month -1 (D1 to D31)

Month -2 (D32 to D59)

Evolution of DWH

Stage :5



- ODS:
 - * Operational Data Store
 - * Used for Near current data reporting
 - * Contains current granularity period data .
 - * ODS data is Detail and Dynamic.